Type 2 diabetes glucose-risk prediction prototype | SRM University AP

Exploratory prediction of future glucose threshold events

Competition prototype overview

SYNTHETIC RESEARCH DEMO — NOT FOR CLINICAL USE

Team: mathin7225 | 2 members

Leader: Shaik Mahammad Mathin (SRM University AP) | Teammate: Shaik Sameera Kowsar (Tirumala Engineering College)

Incubator: None | Video: PENDING genuine unlisted URL (20+ minutes)

Problem and healthcare use case

A software research objective, not a clinical intervention.

Question

Can information available at a prediction time help identify whether Libre glucose will exceed 180 mg/dL in the following two hours?

- Potential use case: explore a future-risk alert concept for research and education.
- Target: any observed Libre value strictly above 180 mg/dL in (t, t + 120 minutes].
- The endpoint requires observed Libre coverage and a maximum five-minute gap; Dexcom does not fill gaps or define labels.
- No care recommendation, diagnosis, or clinical action is implemented.

Synthetic data streams

Static profiles and dynamic signals join by participant ID before modeling.

A | Static synthetic profile

Per participant:
age, sex, diagnosis,
diabetes duration, BMI,
HbA1c

B | Dynamic synthetic fixture

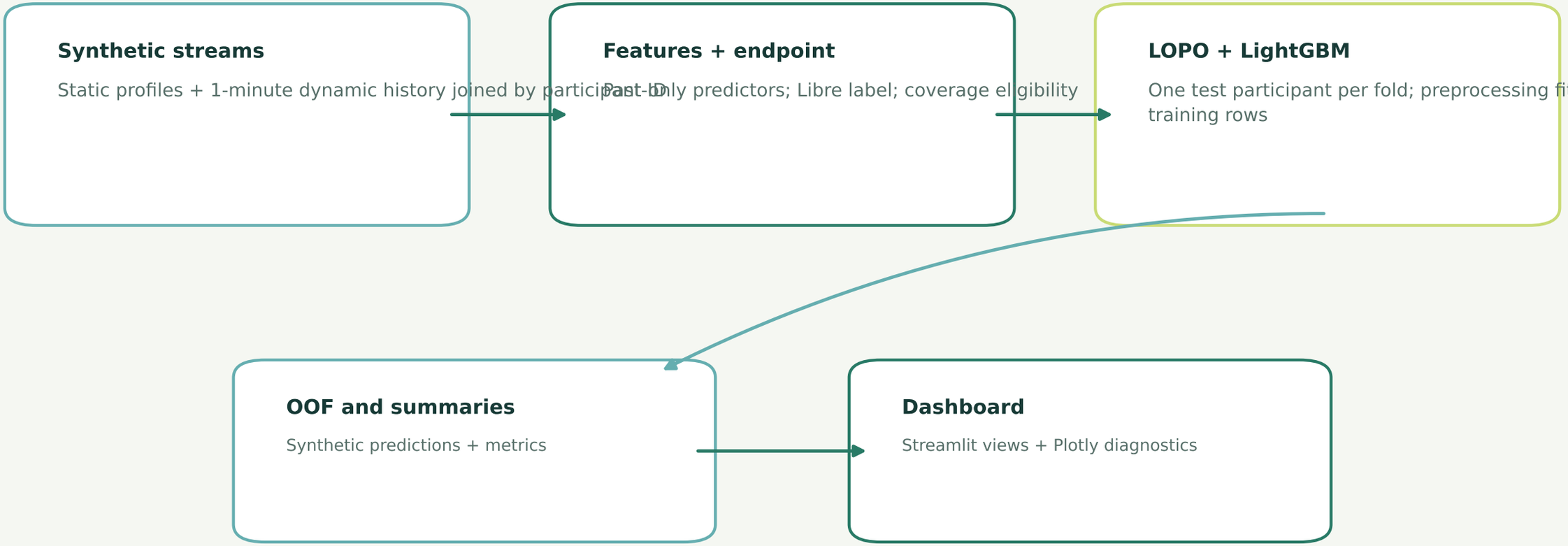
One-minute fixture:
Libre glucose, meals,
heart rate, activity

Static and dynamic rows are joined by synthetic participant ID before causal feature generation and LOPO training.

The separate five-minute CSV series from src/synth_data.py is not used as model input; its static profile helper is reused. Neither stream is patient data or real-time hardware.

Architecture and data flow

The architecture PDF is a separate vector diagram in this folder.



New-scenario inputs use the same ordered 151-feature schema, including recent meal type, and a fresh synthetic-only fit; output is separate from historical OOF replay. No real data or hardware stream enters this demo.

Model and validation

Configuration shown here is the actual deterministic synthetic smoke configuration.

- Model: LightGBM classifier, 75 estimators, learning rate 0.08, max depth 3, 15 leaves.
- Preprocessing: numeric median imputation; categorical most-frequent imputation plus one-hot encoding, fitted within each training fold.
- Split: leave-one-patient-out; one complete synthetic participant held out per fold; anchors every 10 elapsed minutes.
- Prediction cutoff: fixed 0.5; no threshold tuning.
- Metrics: pooled OOF AUROC, average precision (AUPRC), Brier score, prevalence; anchor and event summaries; majority baseline.
- Uncertainty: participant bootstrap is implemented for per-participant anchor/event metrics; pooled probability metrics shown next are point estimates without participant-bootstrap intervals.

Observed synthetic smoke results

Fixed run: seed 42 | 4 synthetic participants | 2 synthetic days | 1,104 eligible anchors

AUROC	AUPRC	Brier	Positive prevalence
0.852	**0.601**	**0.073**	**0.116**

Anchor classification at fixed 0.5

Training-fold majority	Precision 0.000	Recall 0.000	F1 0.000
LightGBM	Precision 0.681	Recall 0.367	F1 0.477

LightGBM event summary: 8 synthetic events; 17 alert episodes; 4 matched; 13 false alert episodes; 1.70 false alerts per measured day.

Same-data ablation without meal_type_last_120m: AUROC 0.845, AUPRC 0.608, Brier 0.072. Mixed results; no overall improvement claimed.

All figures are synthetic software checks. They do not establish clinical validity or utility.

Dashboard demonstration

All views below are rendered from the synthetic smoke bundle.

Overview

Seed, synthetic participant count, eligible anchors, prevalence, model settings, metrics, and baseline comparisons.

OOF Explorer

Filter synthetic out-of-fold predictions by participant and held-out fold.

Model Diagnostics

ROC and precision-recall curves, reliability diagram, fixed-cutoff confusion matrix, probability distribution.

Synthetic Glucose Timeline

Plot the generated synthetic CGM history for a selected synthetic participant.

New Synthetic Scenario

Enter current and timestamped history measurements, add/edit/remove manual meals, then run a separate 120-minute scenario prediction with missing-input disclosure.

Limitations and future work

The prototype demonstrates software behavior, not clinical validation.

- No real CGMacros evaluation is evidenced; current metrics and new-scenario training are synthetic only.
- The new-scenario probability is not calibrated or clinically validated; incomplete history leaves predictors missing and the pipeline imputes them.
- The separate five-minute CSV time series is not consumed by the dashboard; the shared static profile helper and its separate one-minute dynamic fixture are combined by participant ID.
- Dataset use rights, citation, challenge rules, and redistribution requirements need current authoritative verification.
- Source signal acquisition cadence versus upstream interpolation and sensor delivery latency remain unknown; no real-time hardware is connected.
- Pooled AUROC/AUPRC/Brier do not have participant-bootstrap intervals; no fitted calibration or tuned operating point is provided.
- Future: verify rights and cadence; wire versioned data provenance/run manifests; evaluate per-patient eligibility and missingness; prespecify external validation and calibration; follow official submission format.

Submission actions

Complete personal, legal, and publication details before sharing the final package.

- Team: mathin7225, 2 members. Leader: Shaik Mahammad Mathin (SRM University AP); teammate: Shaik Sameera Kowsar (Tirumala Engineering College). Incubator: None, per current project information.
- Video: PENDING genuine unlisted URL; confirm duration is at least 20 minutes and link access works without requesting permission.
- License: MIT names Shaik Mahammad Mathin. The named holder must confirm authority to license all included contributions; third-party data/dependencies retain their own terms.
- Required folder convention: mathin7225_SRM University AP. Do not rename this Git repository. The package is incomplete until the video is supplied.
- Public release: verify the repository and PDF links open in a signed-out browser; do not publish dataset contents or private identifiers.